title: "The Universe as a Somatic Organism: A Formal Derivation of the 11-Dimensional Manifold from First Principles" subtitle: "Fractal Programme — Foundation Paper" author: "Alistair Johnson" orcid: "0009-0007-2194-0850" institute: "Independent Researcher, Zurich, Switzerland" date: "2026" lang: en-GB bibliography: ../../paper/bibliography.bib csl: ../../paper/apa-7th.csl abstract: | We present a formal derivation of an eleven-dimensional field-theoretic architecture from the minimum degrees of freedom required to describe a self-aware physical system. The derivation is inductive: we do not import M-theory and claim isomorphism; we count functional degrees of freedom and discover the isomorphism as a theorem. The governing equation at every scale is the Helmholtz Green's function $(\nabla^2 + k^2)G = \delta$, with the wavenumber k varying across scale and the form of the equation invariant. At the cosmological limit, this identifies the Green's function of spacetime — the gravitational wave propagator — as a special case of the same master equation that governs synaptic transmission at scale 6. We verify the structural claims in Lean 4 and state the five open proof obligations explicitly. This paper is the foundation of the Fractal Programme: subsequent papers apply the same master equation to geophysics, social dynamics, game theory, and law, each written for the relevant domain's audience without requiring familiarity with the others.